

Progress Report 5

Student Name: Reubin Chatta

Student ID: 300394193

Project: Smart Door Anomaly Detection & Cybersecurity for Elderly Care

Course: CSIS 4495 – Applied Research Project, Section 3

Reporting Period: March 17 – March 23, 2026

Report Date: March 23, 2026

Work Date / Hours Logs

Date	Hours	Description of Work Done
Mar 18, 2026	2	Researched YOLOv8 for door security. Created object categories like weapons, parcels, mobility aids, and threats. Updated config, requirements, and gitignore to support object detection.
Mar 19, 2026	3	Built ObjectDetector using YOLOv8n. Mapped COCO classes to categories. Added threat levels (CRITICAL, HIGH, MEDIUM, INFO). Added frame filtering and unattended item timer to reduce false alerts.
Mar 20, 2026	2.5	Created backend APIs for object detection (detect, status, events). Connected them to Flask app and added fallback if model is missing.
Mar 21, 2026	2	Built script to fine-tune YOLOv8 for gun detection. Added training setup and model saving. System loads this model if available.
Mar 23, 2026	2.5	Built frontend dashboard showing detection events, counts, and severity.

		Updated sidebar and API functions. Did report writing and meeting.
--	--	--

Total Hours: 12

Summary Description of Work Done

This week I completed Phase 3: object detection at the door. This system detects security risks and useful objects for elderly care.

Object Categories:

I created 5 groups:

- Weapons → critical alerts
- Parcels → info alerts for deliveries
- Mobility aids → info alerts
- Security threats → high/medium alerts
- Operational hazards → low/medium alerts

Model:

I used YOLOv8n because it runs fast on Raspberry Pi.

I added:

- Frame filtering to avoid false alerts
- Timer to detect unattended items after 2 minutes

Gun Detection:

Since guns are not in COCO dataset, I trained a custom model and added it to the system.

Backend:

I added 3 APIs:

- Detect objects
- Check system status
- Get recent events

All results are saved using existing database system.

Frontend:

I created a dashboard page showing:

- Live events
- Category counts
- Severity levels

Also updated sidebar navigation.

Repo Check-in of Implementation Completed

Implementation/config.py — Object detection settings

Implementation/requirements.txt — Added ultralytics

.gitignore — Ignored model and log files

Implementation/models/object_detection.py — Main detection logic

Implementation/api/object_detection_routes.py — API endpoints

Implementation/api/_init_.py — Setup

Implementation/scripts/finetune_weapon_model.py — Training script

Implementation/frontend/app/objects/page.tsx — Dashboard

Implementation/frontend/components/Sidebar.tsx — Navigation

Implementation/frontend/lib/api.ts — API functions

All changes committed in branch reubinv3.

AI Use Section

AI Tool	Version	Type	Usage	Value
Cursor	Composer/Agent	Code	Detector, API, frontend	Base code generated and customized
ChatGPT	Pro	Writing	Report help	Improved clarity and structure

Appendix: Prompt History

- How to Mapping YOLO classes to categories
- How to Reduce Reducing false positives using frame filtering
- How to Training YOLO for gun detection